

- 15 $\angle AOB$ is a sector of a circle with centre O and radius 12 cm



Diagram NOT
accurately drawn

Work out the area of the sector.

Give your answer correct to 3 significant figures.

$$\begin{aligned}
 \text{Area of sector} &= \frac{\theta}{360} \times \pi r^2 \\
 &= \frac{75}{360} \times \pi \times 12^2 \\
 &= 30\pi \\
 &= 94.247 \\
 &= \underline{\underline{94.2\text{ cm}^2}}
 \end{aligned}$$

94.2 cm²

(Total for Question 15 is 2 marks)

- 16 Ivan asked 15 people how many books they read last year.

Here are his results.

1, 1, 3, 4, 6, 7, 8, 9, 10, 12, 15, 25, 30, 37, 50

Work out the interquartile range of the number of books read.

$$IQR = Q_3 - Q_1$$

Q_3

$$\frac{3}{4}(n+1)$$

$$= \frac{3}{4}(15+1) = 12^{\text{th}} = 25$$

Q_1

$$\frac{1}{4}(n+1)$$

$$= \frac{1}{4}(15+1) = 4^{\text{th}} = 4$$

$$IQR = Q_3 - Q_1$$

$$= 25 - 4$$

$$= \underline{\underline{21}}$$

21

(Total for Question 16 is 2 marks)

